

Analysis group meeting

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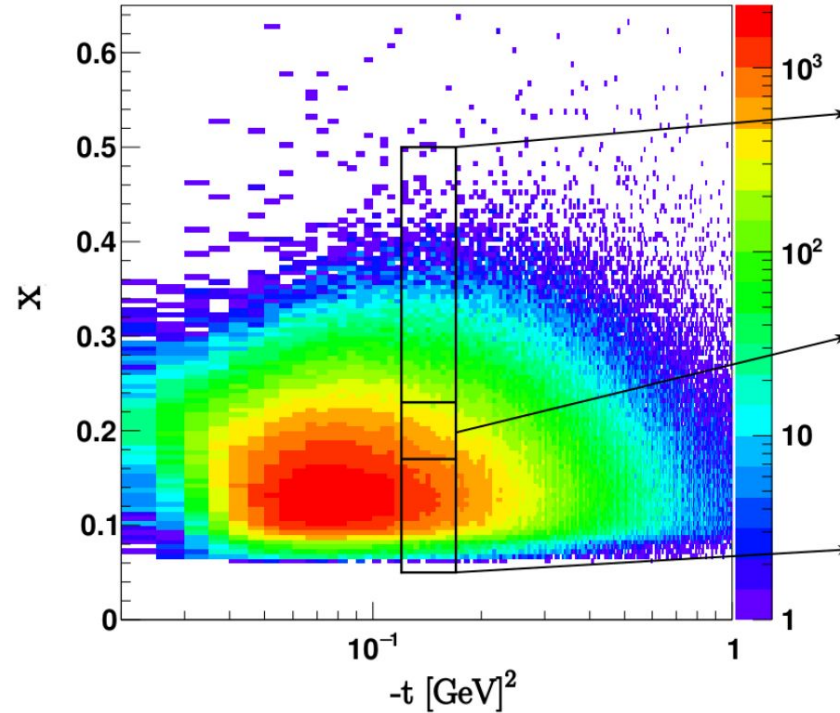
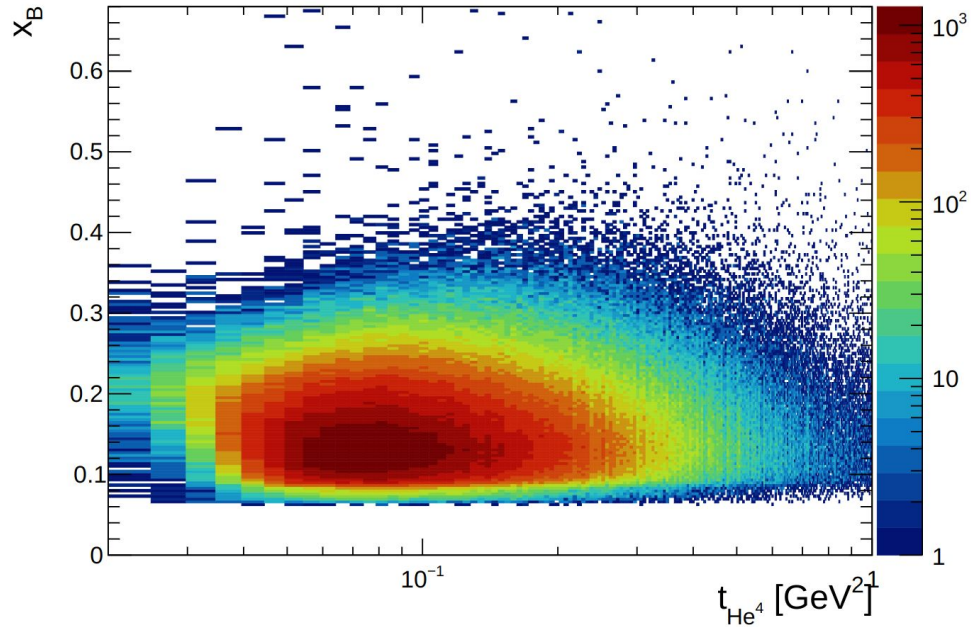


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- Cooking for Pass0v7 and DC calibration is done.
- Took ~5 days (Pass0v7 use 20 files per runs, dc use decoded files from calib_v9)
- Launch new workflow Calib_v10 for CTOF calibration

- Skimed Pass0v7, same location:
`/cache/hallb/scratch/rg-d/production/skim_pass0v7/`

- I have deleted the decoded files for Pass0v7.
- Same for recon files.

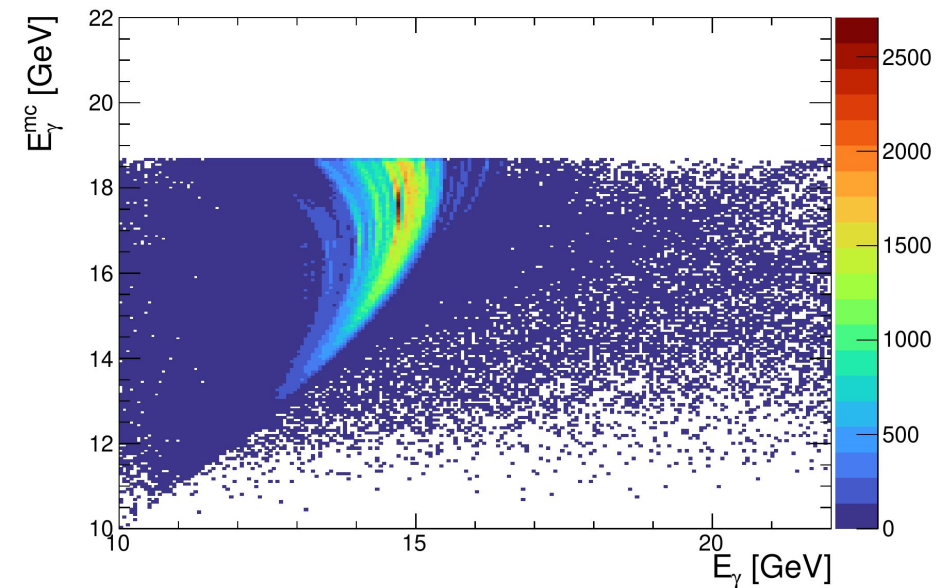


- Not empty banks: REC::Particle, AHDCRec::Track
- Select electron: pid = 11 and status < 0
- Select photon pid = 22 and most energetic + $E > 2$ GeV
- Select Helium4 first track in AHDCRec::Track
 - Why px,py and pz is multiply by 2? Mistake in the reconstruction code (COATJAVA)

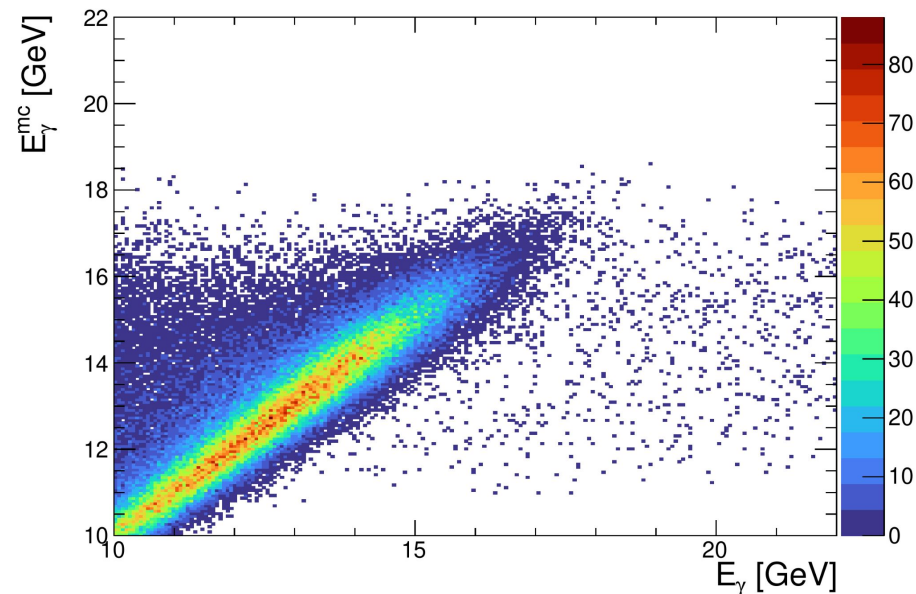
```
v4He.SetXYZM(2.*ahdcrec_track_px[He_ind[0]]/1000.,  
2.*ahdcrec_track_py[He_ind[0]]/1000.,  
2.*ahdcrec_track_pz[He_ind[0]]/1000.,  
hd30::m4he);
```

- Kinematical cuts:
 - $Q^2 > 2.4$ GeV²
- No cuts on θ_e, θ_γ or θ_{He4}

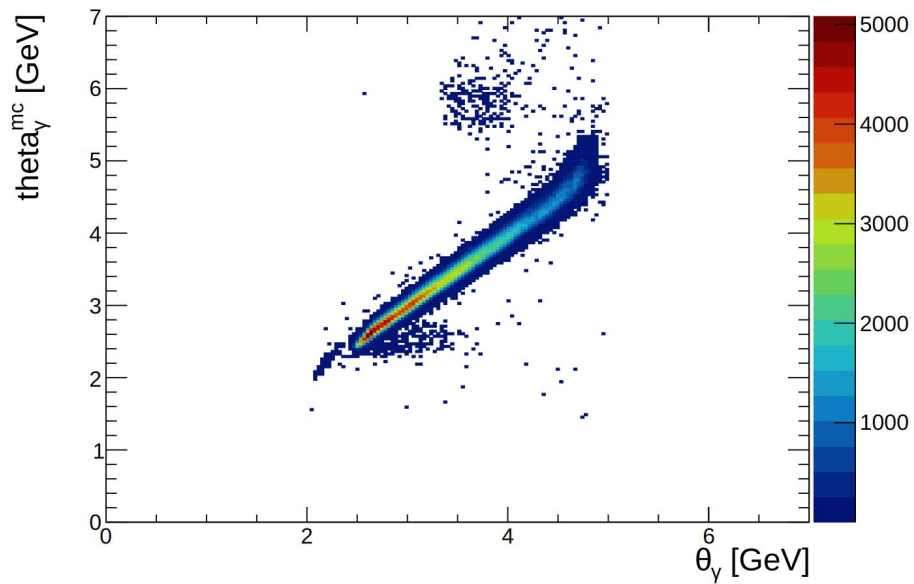
Forward Tagger



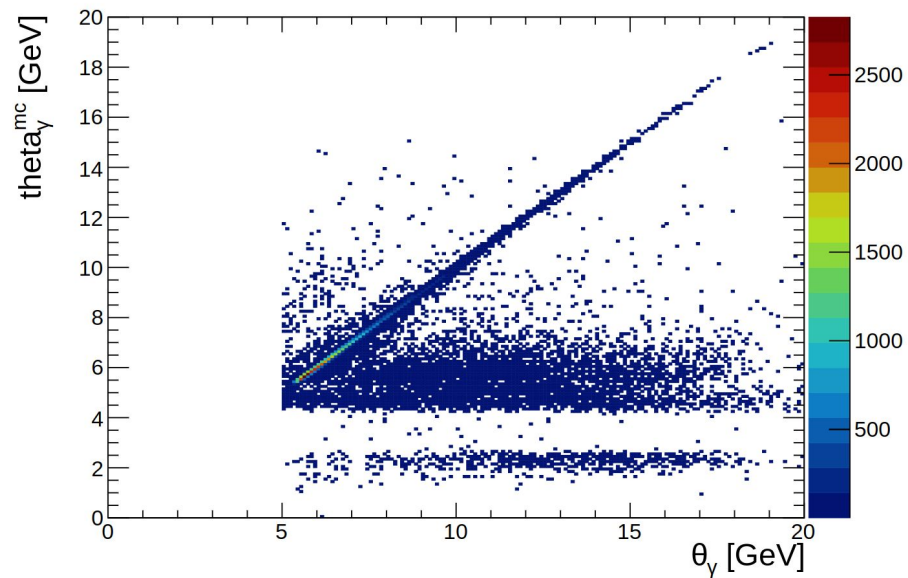
Forward Detector



Forward Tagger



Forward Detector



$$t_{\gamma\gamma} = \frac{-Q^2 - 2\nu_1(\nu_1 - q_1 \cos \theta_{q_1 q_2})}{1 + (\nu_1 - q_1 \cos \theta_{q_1 q_2})/M}$$



involve 2->2 reaction and target at rest
<https://arxiv.org/pdf/1708.00835>

$$\begin{aligned} t_q &= (q_1 - q_2)^2 \\ &= -Q^2 - 2\nu_2(\nu_1 - q_1 \cos \theta_{q_1 q_2}) \end{aligned}$$

$$\begin{aligned} t_p &= (p_1 - p_2)^2 \\ &= 2M^2 - 2(E_1 E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2). \end{aligned}$$

- AI for ALERT:
 - Implement AI in COATJAVA, know how to do it in Java.
- ALERT@22GeV
 - Reproduce Mikhail result
 - Extend it to 2 times luminosity and lower Q^2
 - Incoherent DVCS on He^4
- RG-D:
 - Work on nuclear transparency vs Q^2 and vs l_c
 - Work on K0 analysis, check the event of $z_h > 0.3$ cut, improve K0 identification with detached vertex, compute multiplicity ratio and asymmetry
- Abstract for GHP:
 - Can be on AI for ALERT